LECTURE 11

LAST TIME

LECTURE 10: SUMMARY

Utility maximization

Portfolio choice

Asset pricing and the SDF

CAPM

THIS TIME

LECTURE 11 OUTLINE

More on the FTAP

Introduction to Arrow securities

Market price of risk

Repeated binomial model

WARM-UP EXERCISE

WARM-UP EXERCISE

Binomial model w/ $r = 1$, $S_0 = 1$, $u = 3$, $d = 0$, $p = 1/4$

Find risk-neutral probability $\mathbb{Q}$

Price a call S with strike 2

Price a put on S with strike 1

Relate $\mathbb{Q}[u]$ and $\mathbb{Q}[d]$ to call and put prices?

Relate $\mathbb{Q}[u]$ and $\mathbb{Q}[d]$ to $\mathbb{P}[u]$ and $\mathbb{P}[d]$?

WARM-UP SOLUTION

$$q \equiv \frac{1+r-d}{u-d} = \frac{1+1-0}{3-0} = \frac{2}{3}, \implies 1-q = \frac{1}{3}$$

For call $c_u = 3-2 = 1$, $c_d = 0$, so, by FTAP,

$$\begin{aligned} c_0 &= \frac{qc_u + (1-q)c_d}{1+r} \\ &= \frac{\frac{2}{3}\cdot 1 + \frac{1}{3}\cdot 0}{2} \\ &= \frac{1}{3} \end{aligned}$$

WARM-UP SOLUTION

For the put, $p_u = 0$, $p_d = 1 - 0 = 1$, so, by the FTAP,

$$\begin{aligned} p_0 &= \frac{qp_u + (1-q)p_d}{1+r} \\ &= \frac{\frac{2}{3} \cdot 0 + \frac{1}{3} \cdot 1}{2} \\ &= \frac{1}{6} \end{aligned}$$

Observe $q = (1+r)c_0$ & $1 - q = (1+r)p_0$

Here c and p are Arrow securities: i.e. securities with payoff $X_\omega = \mathbb{1}_{\{\omega\}}$

QUICK QUESTION

Suppose a derivative X pays off S_T in every state, $X_T \equiv S_T$

What is its price today?

What changes if derivative pays off ex dividend?

QUICK SOLUTION

Claim to cash flow S_T is asset $S \implies X_0 = S_0$

If X includes claim to dividends, $X_0 < S_0$. Why?

ANOTHER WAY TO SEE $\mathbb{Q}$

$\mathbb{Q}$ probability such that $S_0 = \dfrac{1}{1+r}\mathbb{E}[S_T]$

In the binomial model this means

$$S_0 = \frac{quS_0 + (1-q)dS_0}{1+r}$$

and rearranging gives

$$q = \frac{1+r-d}{u-d}$$

as before

NB: "Probability" means weighting/density; price weighted average

ARROW SECURITIES EXERCISE

ARROW SECURITIES EXERCISE

Consider binomial model with $S_0 = 2$, $S_u = 3$, $S_d = 1$, $r = 0$

1. Find risk-neutral probability $\mathbb{Q}$
2. Find prices of Arrow securities

   I.e. find ϕ^ω, price of claim to $\mathbb{1}_{\{\omega\}}$ for $\omega \in \{u, d\}$
3. Find portfolios that replicate Arrow securities:

   (β_u, Δ_u) with payoff $\mathbb{1}_{\{u\}}$

   (β_d, Δ_d) with payoff $\mathbb{1}_{\{d\}}$

ARROW SECURITIES SOLUTION

Calculate $u = 3/2$, $d = 1/2$, and recall $r = 0$ so

$$q = \frac{1 + 0 - 1/2}{3/2 - 1/2} = \frac{1}{2}$$

FTAP: $\phi^u = \frac{1}{1+r} q = \frac{1}{2}$ and $\phi^d = \frac{1}{1+r}(1-q) = \frac{1}{2}$

ARROW SECURITIES SOLUTION

Calculate $u = 3/2$, $d = 1/2$, and recall $r = 0$ so

$$q = \frac{1 + 0 - 1/2}{3/2 - 1/2} = \frac{1}{2}$$

FTAP: $\phi^u = \frac{1}{1+r} q = \frac{1}{2}$ and $\phi^d = \frac{1}{1+r}(1-q) = \frac{1}{2}$

How could you have guessed this solution without calculations?

TRINOMIAL WARM-UP

TRINOMIAL WARM-UP

Consider trinomial (three-state) model with two securities with payoffs

$$B_T = \begin{pmatrix} 100 \\ 100 \\ 100 \end{pmatrix} \quad \text{and} \quad S_T = \begin{pmatrix} 256 \\ 144 \\ 81 \end{pmatrix}$$

and prices $B_0 = 100$ and $S_0 = 144$

1. Find the forward price $F_0(T)$ of S

2. Can you replicate a call on S_T with strike $K = 200$

   If so, what is the price c_0? If not, why?

3. Suppose $c_0 = 72/7$. Can you price a put with strike $K' = 130$?

   If so, what is the price p_0? If not, why?

REPEATED (RECOMBINING) BINOMIAL TREE

REPEATED BINOMIAL TREE

Three dates, Date 0, Date 1, and Date 2

At Date 2, three states (terminal nodes), $\omega \in \{uu, ud, dd\}$

called recombining when $ud = du$

At Date 1 there are two nodes $u = \{uu, ud\}$ and $d = \{ud, dd\}$

EXAMPLE: OPTION PRICING ON REPEATED TREE

OPTION PRICING EXAMPLE

Two assets, a stock S and a bond B

Risk-free rate is zero: $B_t \equiv 1$

From Date t to Date $t+1$, S goes up by $u = 4/3$ or down by $d = 3/4$

S has price $S_0 = 144$ today

Price a call on S that matures at Date 2 with strike $K = 200$

OPTION PRICING STEPS

Divide the tree into three binomial subtrees

Use the FTAP on the subtrees to find c_u and c_d

Use the FTAP to find c_0 from c_u and c_d

NB: could find replicating portfolios at each date instead

THE TREE

Since $r = 0$, $B_0 = B_u = B_d = B_{uu} = B_{ud} = B_{dd} = 1$

Find S at each node:

$$
\begin{aligned}
S_0 &= 144, \\
S_u &= 144 \cdot \frac{4}{3} = 192, \\
S_d &= 144 \cdot \frac{3}{4} = 108, \\
S_{uu} &= 192 \cdot \frac{4}{3} = 256, \\
S_{ud} &= 192 \cdot \frac{3}{4} = 108 \cdot \frac{4}{3} = 144 \\
S_{dd} &= 108 \cdot \frac{3}{4} = 81
\end{aligned}
$$

$K = 200$ implies $c_{uu} = 56$, $c_{ud} = c_{dd} = 0$

RISK-NEUTRAL PROBABILITIES

First find q:

$$\begin{aligned} q &= \frac{1+r-d}{u-d} \\ &= \frac{1+0-\frac{3}{4}}{\frac{4}{3}-\frac{3}{4}} \\ &= \frac{3}{7} \end{aligned}$$

So

$$1-q = \frac{4}{7}$$

STATE-u PRICE

FTAP:

$$\begin{aligned} c_u &= \frac{1}{1+r}\big(qc_{uu} + (1-q)c_{ud}\big) \\ &= \frac{3}{7} \cdot 56 \\ &= 24 \end{aligned}$$

STATE-d PRICE

Since $c_{ud} = c_{dd} = 0$, in state d the call is worthless

Thus, $c_d = 0$

DATE-0 PRICE

FTAP:

$$\begin{aligned} c_u &= \frac{1}{1+r}\big(qc_u + (1-q)c_d\big) \\ &= \frac{3}{7}\cdot 24 \\ &= \frac{72}{7} \\ &\approx 10.3. \end{aligned}$$

REPEATED TREE EXERCISE

Price a put on the same asset with strike $K' = 130$

If the put were American, would the price change?

OPTION PAYOFFS

First, write the option payoffs at the terminal nodes

$$p_{uu} = (130 - 256)^+ = 0,$$
$$p_{ud} = (130 - 144)^+ = 0,$$
$$p_{dd} = (130 - 81)^+ = 49$$

STATE-u PRICE

Since $p_{uu} = p_{ud} = 0$, in state u the put is worthless

Thus, $p_u = 0$

STATE-d PRICE

FTAP:

$$\begin{aligned} p_d &= \frac{1}{1+r}\big(qp_{ud} + (1-q)p_{dd}\big) \\ &= \frac{4}{7} \cdot 49 \\ &= 28 \end{aligned}$$

DATE-0 PRICE

FTAP:

$$\begin{aligned} p_0 &= \frac{1}{1+r}\big(qp_u + (1-q)p_d\big) \\ &= \frac{4}{7} \cdot 28 \\ &= 16 \end{aligned}$$

WHAT IF THE PUT WERE AMERICAN?

If the put were American, you could exercise early

Check at u and d whether it's better to exercise or not

If you exercise at u, you get

$$(K' - S_u)^+ = (130 - 192)^+ = 0$$

Should you exercise?

WHAT IF THE PUT WERE AMERICAN?

If you exercise at d, you get

$$(K' - S_d)^+ = (130 - 108)^+ = 22$$

Should you exercise?

If you hold the option, it's worth $p_d = 28$

Don't exercise

How could you have guessed this?

Hint: this is a property of this tree, not particular to strike

RECALL: TIME VALUE DECOMPOSITION

Recall the decomposition

$$\text{option price} \;=\; \text{intrinsic value} \;+\; \text{time value}$$

where

$$\text{time value} \;=\; \text{deferred payment value} \;+\; \text{potentiality value}$$

A put is convex, so potentiality value is always positive

But you might exercise early if deferred payment value is positive

Can that happen here?

“DYNAMIC” RISK NEUTRAL PRICING

"DYNAMIC" RISK NEUTRAL PRICING

Note that

$$q^2 c_{uu} = \frac{9}{49} \cdot 56 = \frac{72}{7} = c_0$$

and

$$(1-q)^2 p_{dd} = \frac{16}{49} \cdot 49 = 16$$

In general, can price under $\mathbb{Q}$:

$$\mathbb{Q}[uu] = q^2, \quad \mathbb{Q}[ud] = 2q(1-q), \quad \mathbb{Q}[dd] = (1-q)^2$$

"DYNAMIC" RISK NEUTRAL PRICING EXERCISE

Consider two-step tree with $S_0 = 100$, $u = 2$, $d = 1/2$, and $r = 0$

1. Find the risk-neutral probabilities
2. Price a call on the stock with strike $K = 238$ and maturity 2
3. Price a put on the stock with strike $K' = 112$ and maturity 2

RECAP

LECTURE 12: RECAP

Repeated binomial tree

Backward induction

Lil bout US options (early exercise)

"Dynamic" risk-neutral pricing

FIN

APPENDIX

ARROW SECURITIES SOLUTION

For $\mathbb{1}_{\{u\}}$, replicating equations:

$$\begin{aligned} \beta_u + 3\Delta_u &= 1 \\ \beta_u + \Delta_u &= 0, \end{aligned}$$

so $\Delta_u = 1/2$ and $\beta_u = -1/2$

For $\mathbb{1}_{\{d\}}$, the replicating equations are

$$\begin{aligned} \beta_d + 3\Delta_d &= 0 \\ \beta_d + \Delta_d &= 1, \end{aligned}$$

so $\Delta_d = -1/2$ and $\beta_d = 1 + 1/2 = 3/2$

ARROW SECURITIES SOLUTION

The price of the u-Arrow security is

$$\begin{aligned}\phi^u &= \frac{\beta_u}{1+r} + \Delta_u S_0 \\ &= -\frac{1}{2} + \frac{1}{2} \cdot 2 \\ &= \frac{1}{2}\end{aligned}$$

The price of the d-Arrow security is

$$\begin{aligned}\phi^d &= \frac{\beta_d}{1+r} + \Delta_d S_0 \\ &= \frac{3}{2} - \frac{1}{2} \cdot 2 \\ &= \frac{1}{2}\end{aligned}$$

Can you think of a quicker way to find ϕ^d?

EXAMPLE: PRICE OF RISK IN CAPM WORLD

EXAMPLE: PRICE OF RISK IN CAPM WORLD

Suppose CAPM holds with $\mathbb{E}[R_M] = 2$ and $r = 0$

Consider asset S on binomial tree with $S_u = 4$, $S_d = 0$, $p = 1/2$

The CAPM beta of S is one

Consider two options on S

(i) Call with strike $K = 2$

(ii) Put with strike $K = 2$

Which option do you expect to be dearer?

EXPECTED PRICES

Note the options have the same expected value

$$\begin{aligned}\mathbb{E}[c_T] &= \frac{1}{2}(S_u - K)^+ + \frac{1}{2}(S_d - K)^+ \\ &= \frac{1}{2}(4-2)^+ + \frac{1}{2}(0-2)^+ \\ &= 1\end{aligned}$$

and

$$\begin{aligned}\mathbb{E}[p_T] &= \frac{1}{2}(K - S_u)^+ + \frac{1}{2}(S_d - K)^2 = 1 \\ &= \frac{1}{2}(2-4)^+ + \frac{1}{2}(2-0)^+ \\ &= 1\end{aligned}$$

So which is dearer?

THREE WAYS TO PRICE

We can price three ways

1. FTAP
2. replication
3. CAPM

THREE WAYS TO PRICE

We can price three ways

1. FTAP
2. replication
3. CAPM

1 and 2 always give the same answer; 3 does not. Why?

THREE WAYS TO PRICE

We can price three ways

1. FTAP
2. replication
3. CAPM

1 and 2 always give the same answer; 3 does not. Why?

If we get different answers, which should we believe?

COMPUTE S_0

To replicate or find $\mathbb{Q}$, first find S_0 from CAPM

$$\mathbb{E}\left[\frac{S_T}{S_0}\right] - (1+r) = \beta_S\big(\mathbb{E}\left[R_M\right] - (1+r)\big)$$

expectations under $\mathbb{P}$ or under $\mathbb{Q}$?

Thus,

$$\frac{\frac{1}{2}\cdot 4 + \frac{1}{2}\cdot 0}{S_0} - 1 = 1\cdot\big(2-1\big)$$

or

$$S_0 = 1$$

RISK-NEUTAL PROBABILITIES

First find u and d:

$$u = \frac{S_u}{S_0} = 4 \quad \text{and} \quad d = \frac{S_d}{S_0} = 0$$

Now

$$q = \frac{1 + r - d}{u - d} = \frac{1}{4}$$

OPTIONS PRICES FROM THE FTAP

FTAP:

$$c_0 = \frac{qc_u + (1-q)c_d}{1+r} = \frac{1}{4} \cdot 2 + \frac{3}{4} \cdot 0 = \frac{1}{2}$$

and

$$p_0 = \frac{qp_u + (1-q)p_d}{1+r} = \frac{1}{4} \cdot 0 + \frac{3}{4} \cdot 2 = \frac{3}{2}$$

PRICES FROM REPLICATION: CALL

Replicate the call $c_T = S_T/4$

$$b_c + \Delta_c S_u = c_u,$$
$$b_c + \Delta_c S_d = c_d$$

or

$$b_c + 4\Delta_c = 2,$$
$$b_c = 0.$$

$\Delta_c = 1/2$ and $\beta_c = 0$, so

$$c_0 = \frac{S_0}{2} = \frac{1}{2}$$

PRICES FROM REPLICATION: PUT

Replicate the call $c_T = S_T/4$

$$\begin{aligned} b_p + \Delta_p S_u &= p_u, \\ b_p + \Delta_p S_d &= p_d \end{aligned}$$

or

$$\begin{aligned} b_p + 4\Delta_p &= 0, \\ b_p &= 2. \end{aligned}$$

$b_p = 2$ and $\Delta_p = -b_p/4 = -1/2$, so

$$\begin{aligned} p_0 &= \beta_p + \Delta_p S_0 \\ &= 2 - \frac{1}{2} \cdot 1 \\ &= \frac{3}{2} \end{aligned}$$

PRICES FROM CAPM

Compute the betas of the call and put and price

To find the betas, use the replicating portfolios

Recall multilinearity of Cov: $\mathrm{Cov}[X+Y,Z] = \mathrm{Cov}[X,Z] + \mathrm{Cov}[Y,Z]$

For portfolio $\theta = (b, \Delta)$, the CAMP beta β_θ is

$$\beta_\theta = \frac{\Delta S_0}{bB_0 + \Delta S_0}\beta_S$$

CALL AND PUT BETAS

We can now compute the call and put betas from Δ_c and Δ_p

Recall $b_c = 0$, $\Delta_c = 1/2$ and $b_p = 2$, $\Delta_p = -1/2$

Since $\beta_S = 1$, the options' CAMP betas are thus

$$\begin{aligned}\beta_c &= \frac{\Delta_c S_0}{b_c + \Delta_c S_0}\beta_S \\ &= \frac{\frac{1}{2}\cdot 1}{0 + \frac{1}{2}\cdot 1}\cdot 1 \\ &= 1\end{aligned}$$

and

$$\begin{aligned}\beta_p &= \frac{\Delta_p S_0}{b_p + \Delta_p S_0}\beta_S \\ &= -\frac{1}{3}\end{aligned}$$

CALL AND PUT IN CAPM

Recall $\mathbb{E}[c_T] = \mathbb{E}[p_T] = 1$

For the call, CAPM says

$$\mathbb{E}\left[\frac{c_T}{c_0}\right] - (1+r) = \beta_c\big([R_M] - (1+r)\big)$$

or

$$\frac{1}{c_0} - 1 = 1 \cdot \big(2 - 1\big)$$

so $c_0 = 1/2$.

For the put, CAPM says

$$\mathbb{E}\left[\frac{p_T}{p_0}\right] - R_f = \beta_p\big(\mathbb{E}[R_M] - R_f\big)$$

or

$$\frac{1}{p_0} - 1 = -\frac{1}{3} \cdot \big(2 - 1\big)$$

so $p_0 = 3/2$

PRICE OF RISK IN CAPM WORLD: MORAL

Can always price by replication $\equiv$ under $\mathbb{Q}$

Can use $\mathbb{P}$ and price with CAPM

CAPM beta takes into account price of risk (adjustment from $\mathbb{P}$ to $\mathbb{Q}$)

But CAPM only one-factor model (which one?)

FTAP is fully general—model independent (just no arbitrage)

NB: Models with more factors (cf. so-called APT)

Size, value/growth, momentum,... But subject to same critique